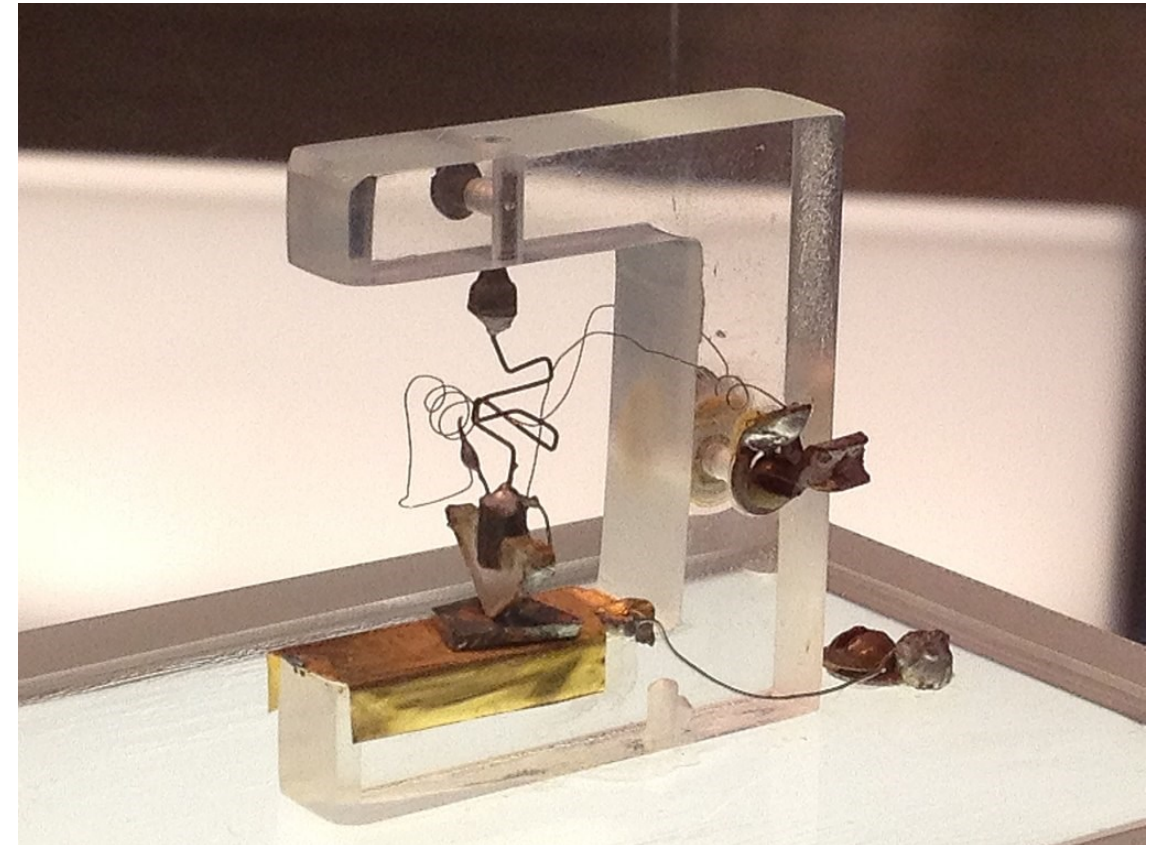
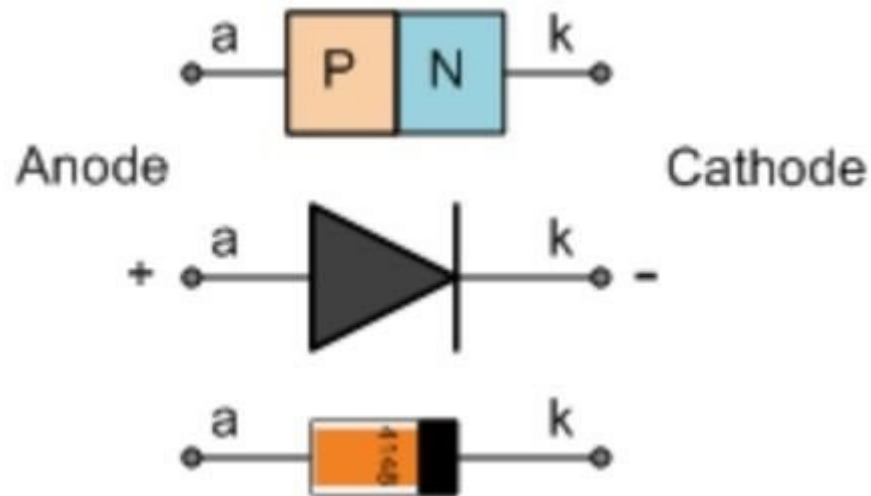


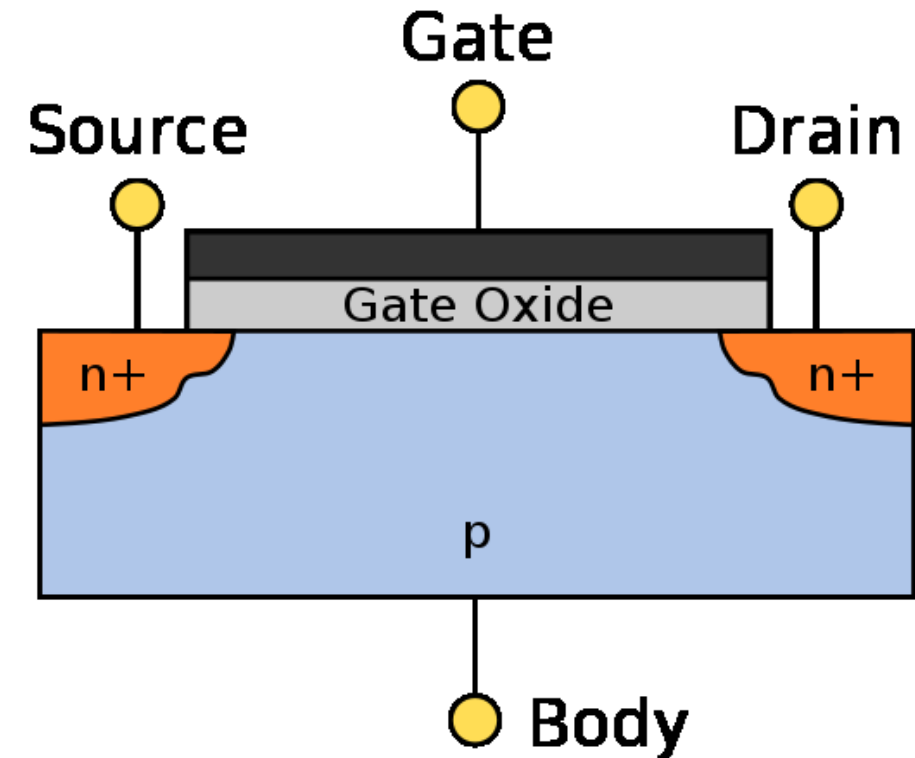
- Diodes allow electrical current to pass only in one direction, in the general ideal case
- In this way, we have a way to create two states by changing the polarity of the provided voltage
- However, we would like a component that allows for two states without the need to change the power source's polarity
- Then, we could create two states through the use of this element
- This demand can be met by a transistor



- While a diode is a two-terminal device, a transistor is a three-terminal one, whose middle terminal allows for controlling the current that goes through it

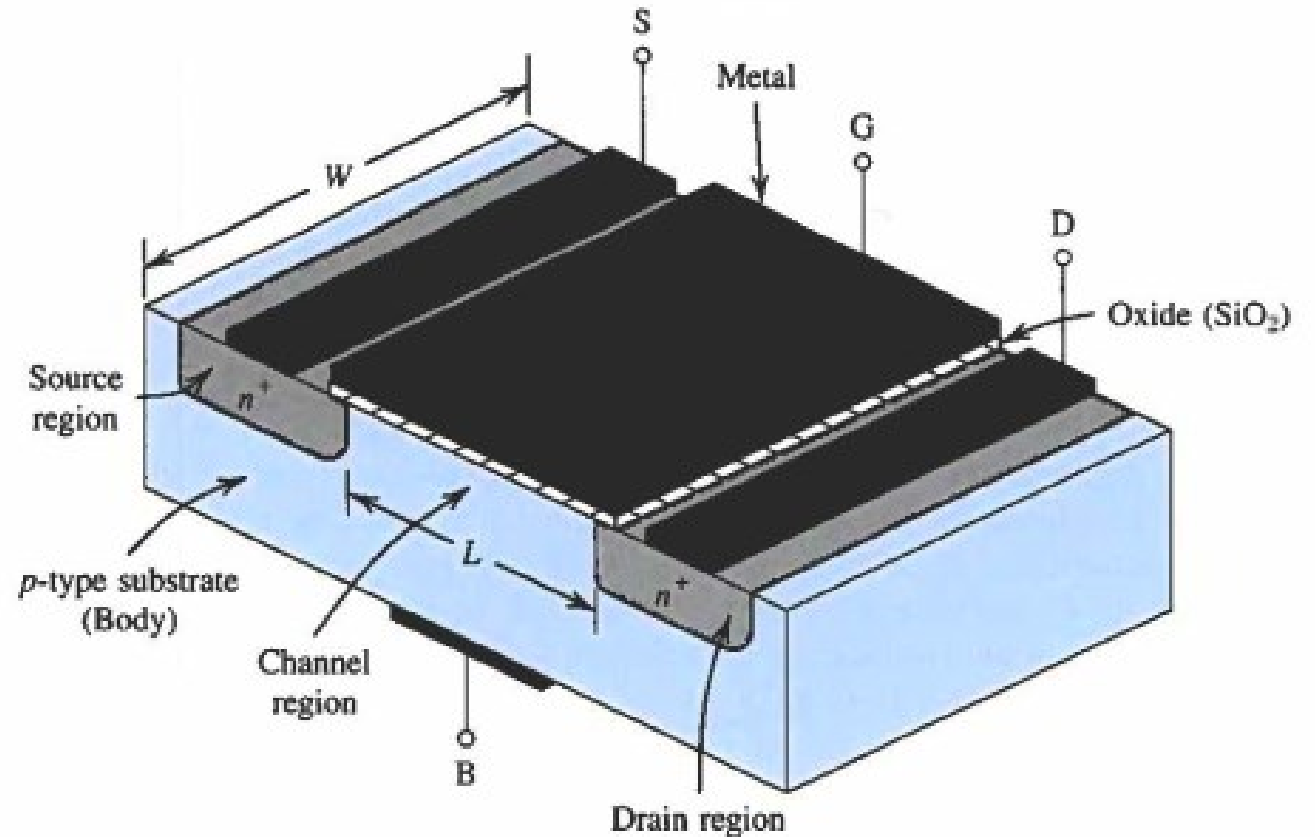
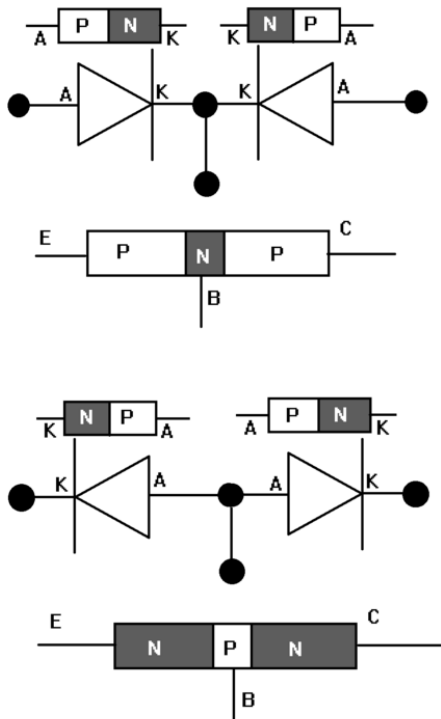


<https://qph.cf2.quoracdn.net/main-qimg-7a67c6ff3c5145927a5ce3c81c3d5ff7-lq>



https://commons.wikimedia.org/wiki/File:FET_cross_section.svg

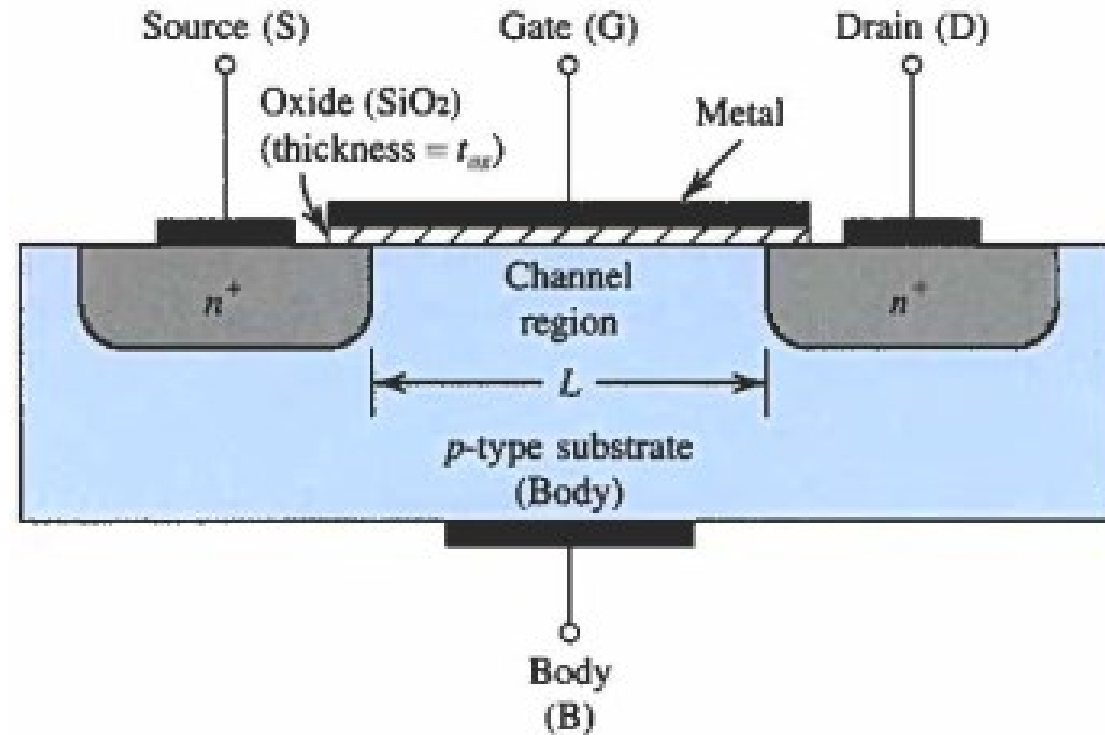
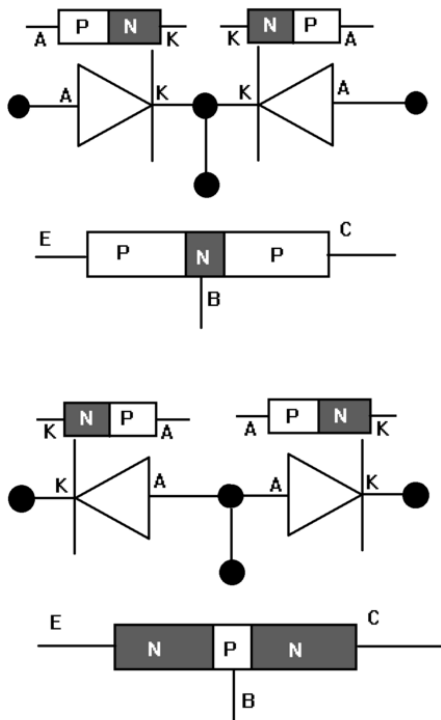
- Metal-Oxide-Semiconductor (MOS)



Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

https://commons.wikimedia.org/wiki/File:Transistor_diode_model.png

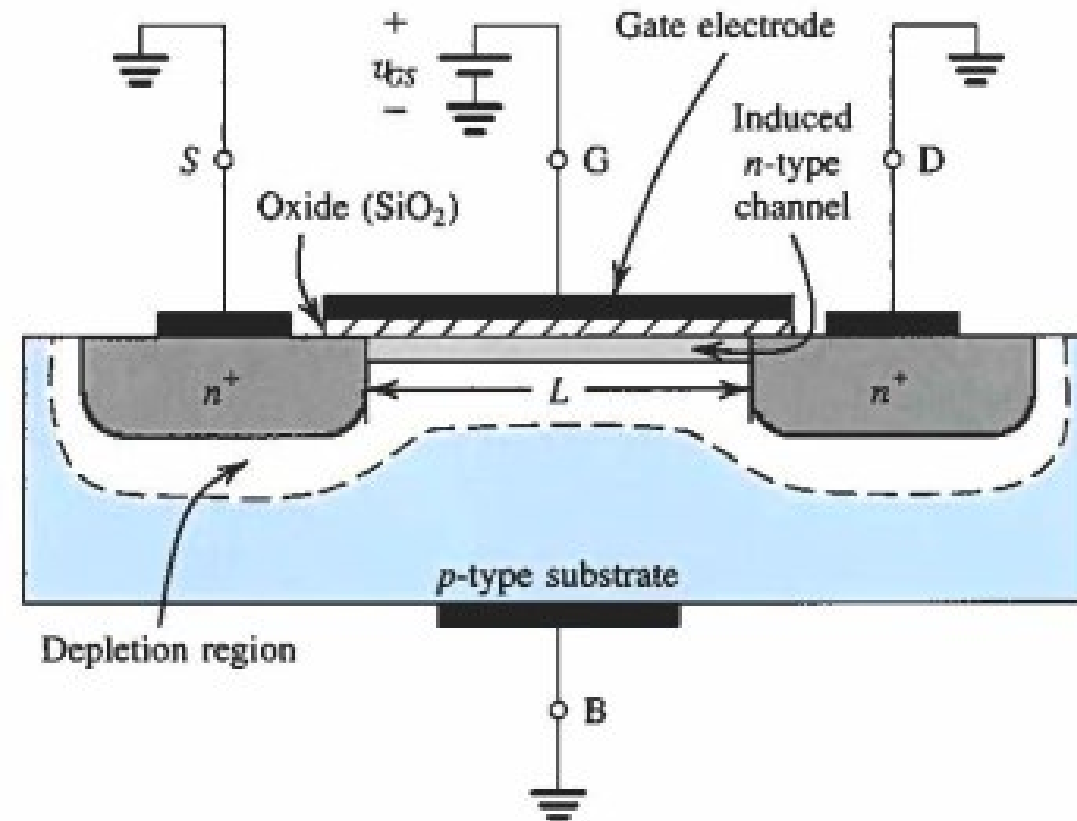
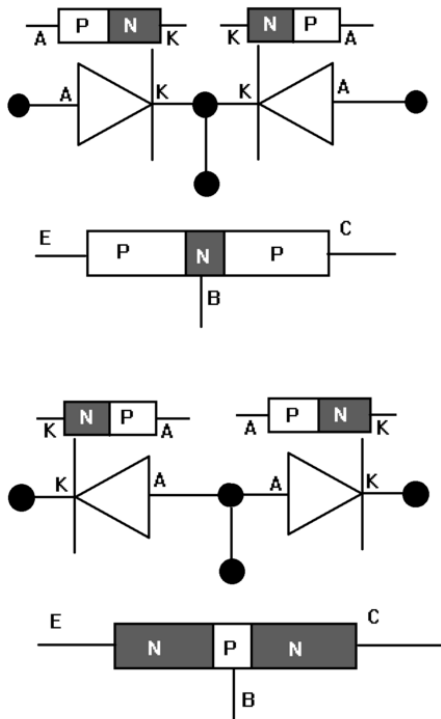
- Metal-Oxide-Semiconductor (MOS)



Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

https://commons.wikimedia.org/wiki/File:Transistor_diode_model.png

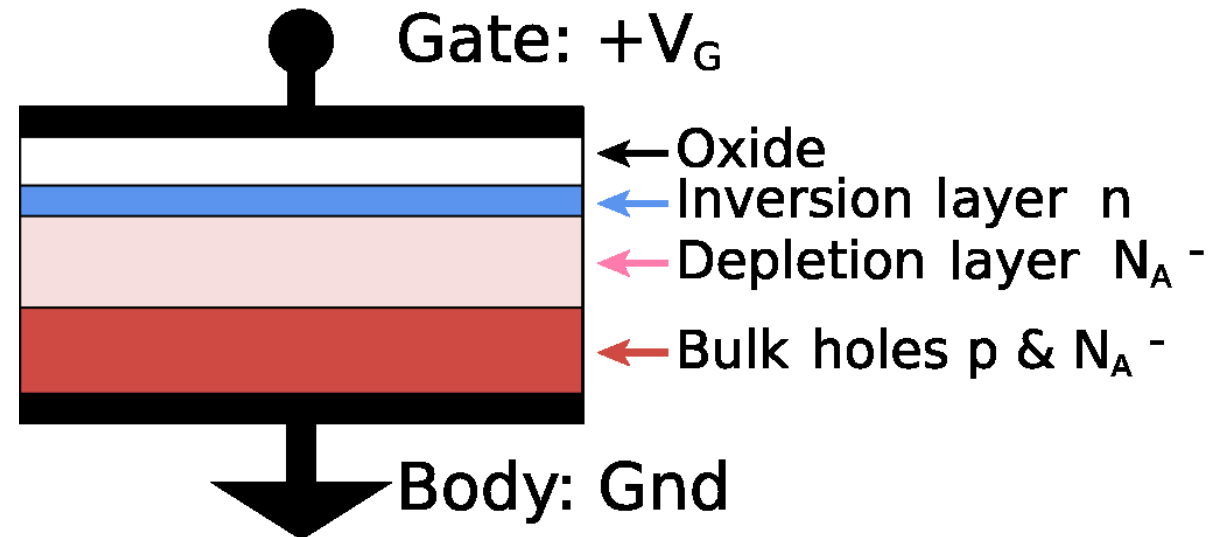
- Metal-Oxide-Semiconductor (MOS)



Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

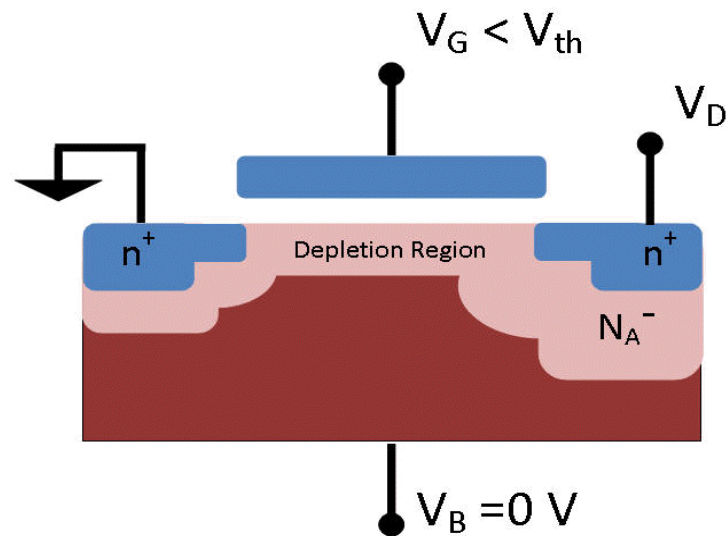
https://commons.wikimedia.org/wiki/File:Transistor_diode_model.png

- Metal-Oxide-Semiconductor (MOS)
- Essentially, a capacitor forms below the gate

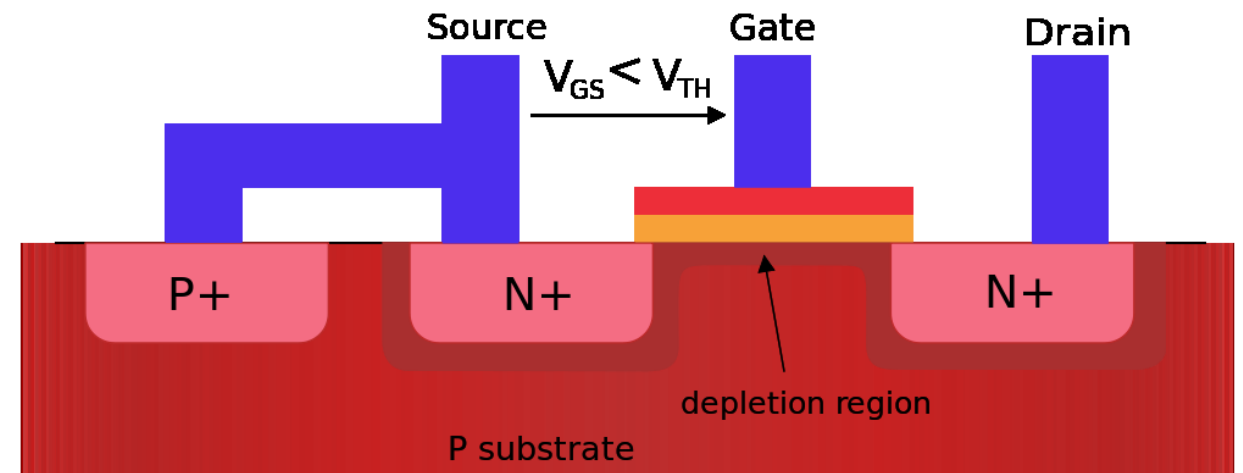


https://en.wikipedia.org/wiki/File:MOS_Capacitor.svg

- The threshold voltage (V_{TH}) is the minimum gate-to-source voltage (V_{GS}) that is needed to create a conducting path between the source and the drain

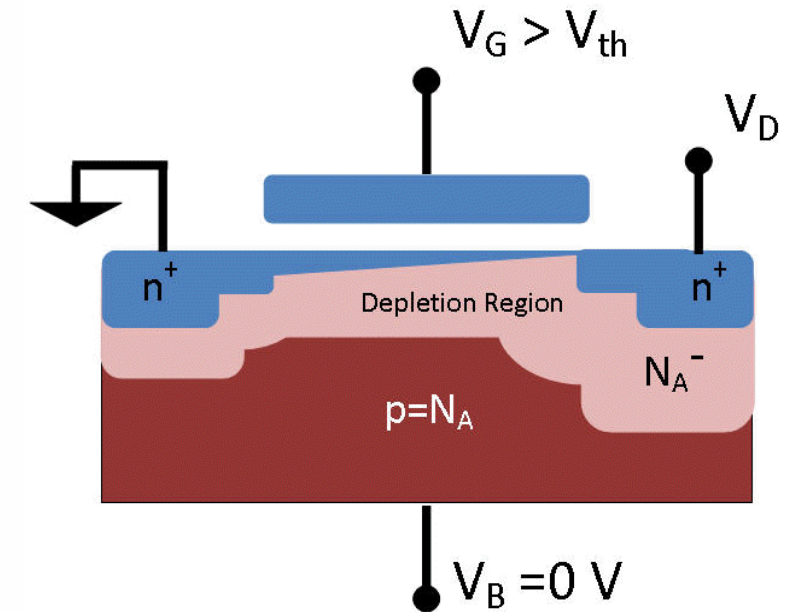
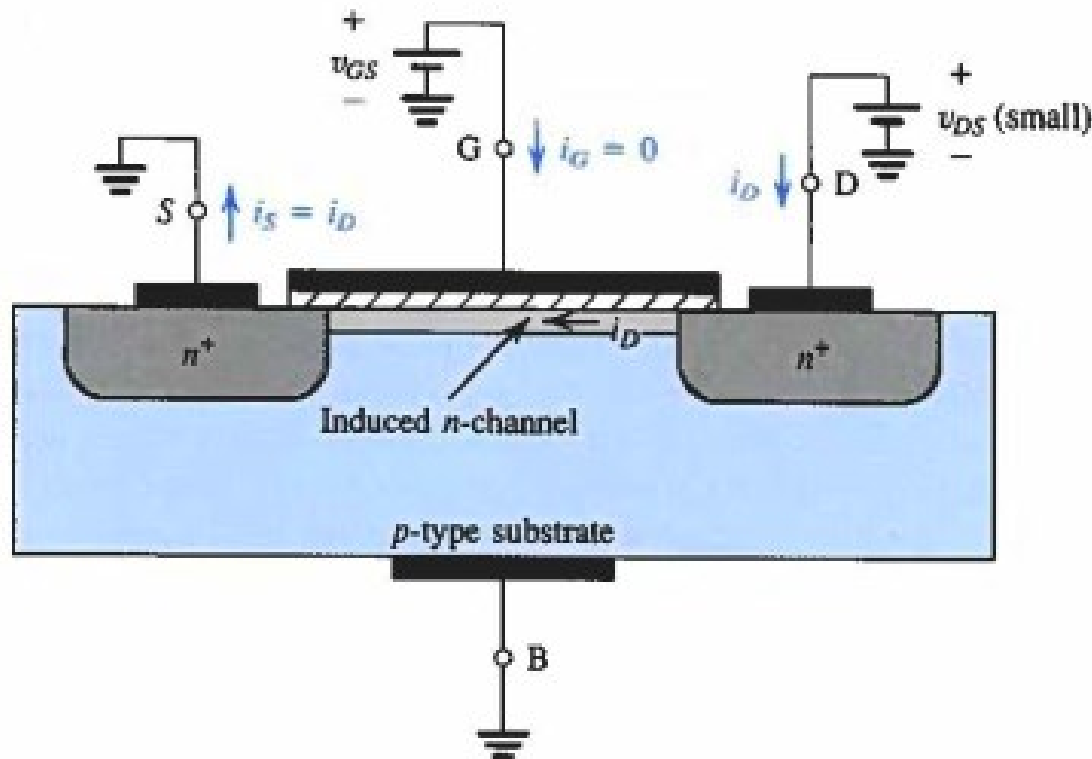


https://commons.wikimedia.org/wiki/File:MOSFET_DepletionRegion.png



https://en.wikipedia.org/wiki/File:MOSFET_functioning_body.svg

Operation of the MOS Transistor



https://commons.wikimedia.org/wiki/File:MOSFET_above_threshold.png

Figure 5.3 An NMOS transistor with $v_{GS} > V_t$ and with a small v_{DS} applied. The device acts as a resistance whose value is determined by v_{GS} . Specifically, the channel conductance is proportional to $v_{GS} - V_t$, and thus i_D is proportional to $(v_{GS} - V_t)v_{DS}$. Note that the depletion region is not shown (for simplicity).

Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

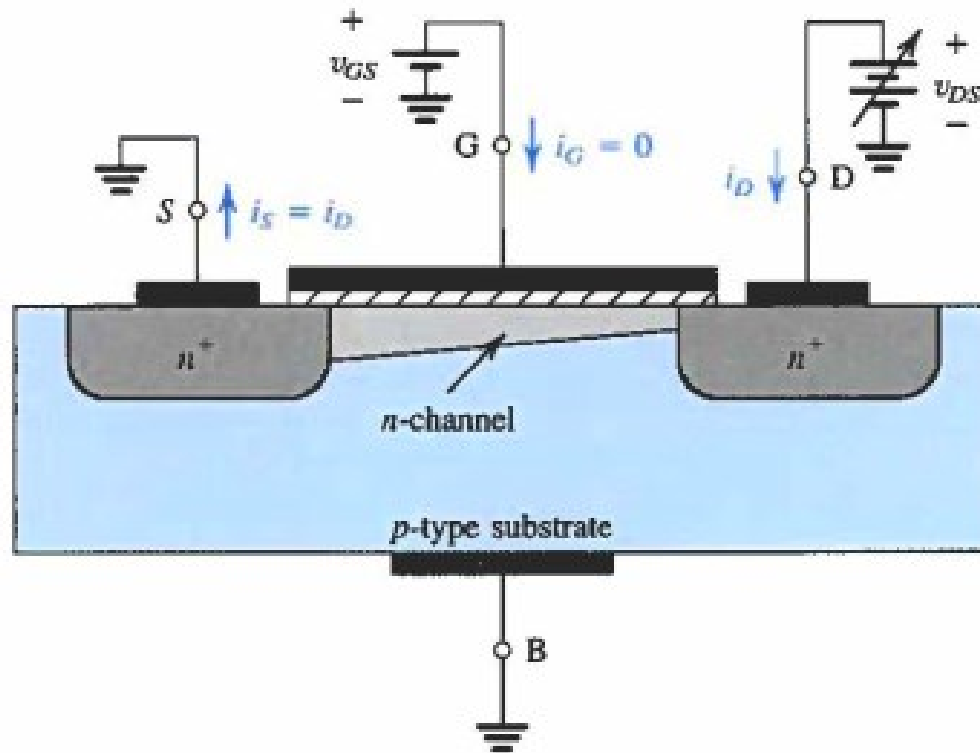
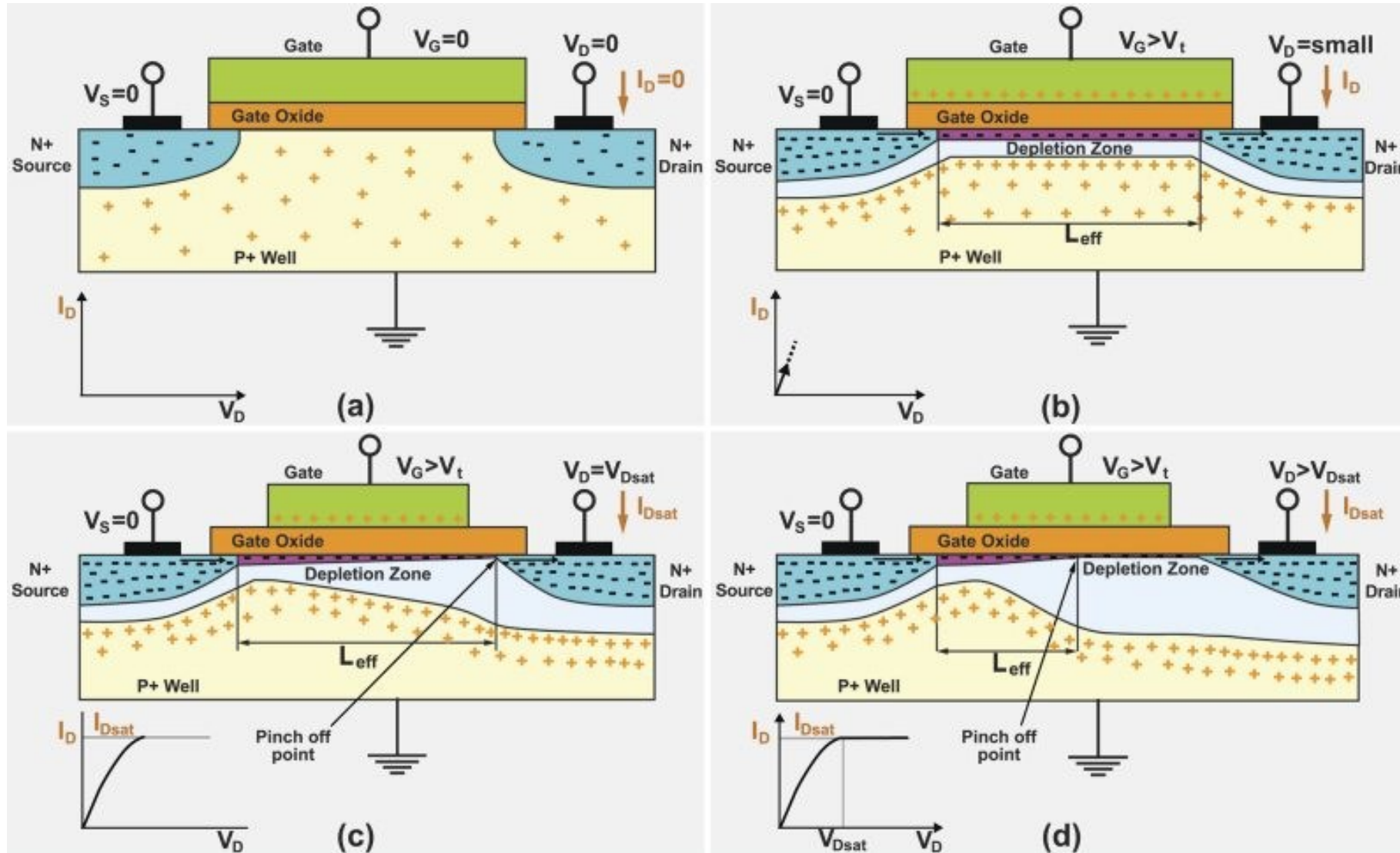


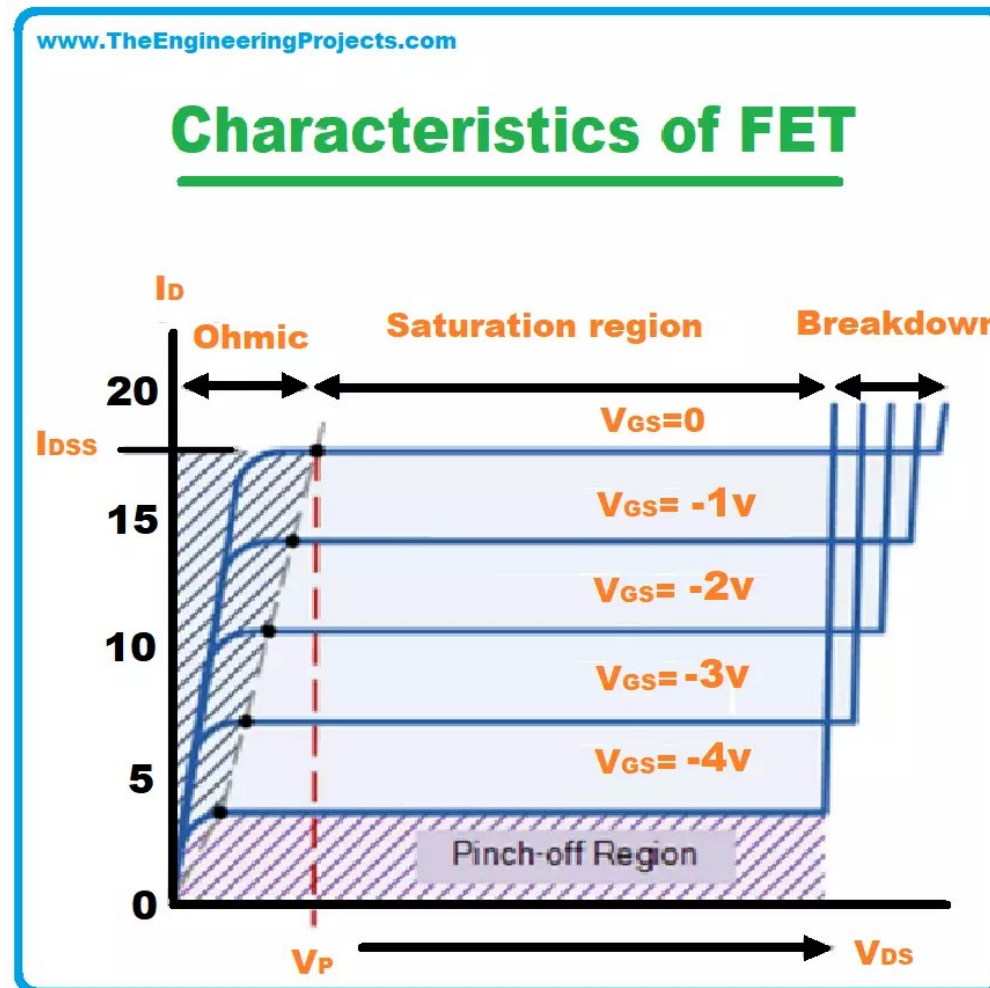
Figure 5.5 Operation of the MOSFET as v_{DS} is increased. The induced channel acquires a tapered shape, and its resistance increases as v_{DS} is increased. Here, v_{GS} is kept constant at a value $> V_t$; $v_{GS} = V_t + v_{OV}$. It is the widening of the depletion region (not shown) as a result of the increased v_{DS} that makes the channel shallower near the drain.

Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

Operation of the MOS Transistor

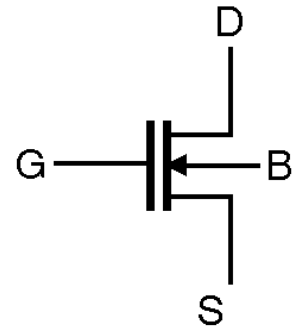


https://www.mks.com/mam/celum/celum_assets/Figure_13-Semiconductor_Handbook_800w.jpg

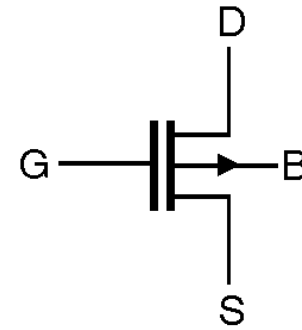


<https://images.theengineeringprojects.com/image/webp/2021/07/Characteristics-of-FET.png.webp?ssl=1>

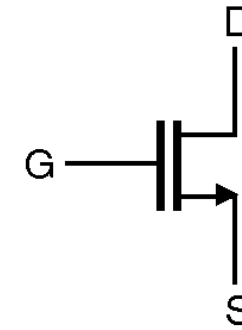
MOS Transistor Types: nMOS & pMOS



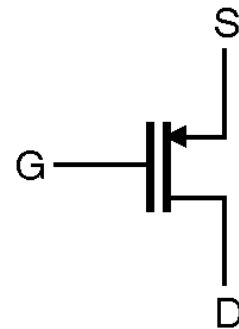
(a)
**NMOS transistor
as four terminal
device**



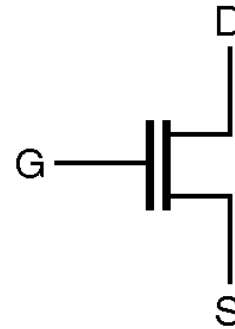
(b)
**PMOS transistor
as four terminal
device**



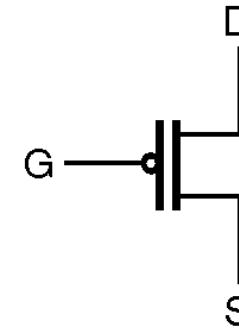
(c)
**NMOS used in
Analog circuits**



(d)
**PMOS used in
Analog circuits**



(e)
**NMOS used in
Digital circuits**

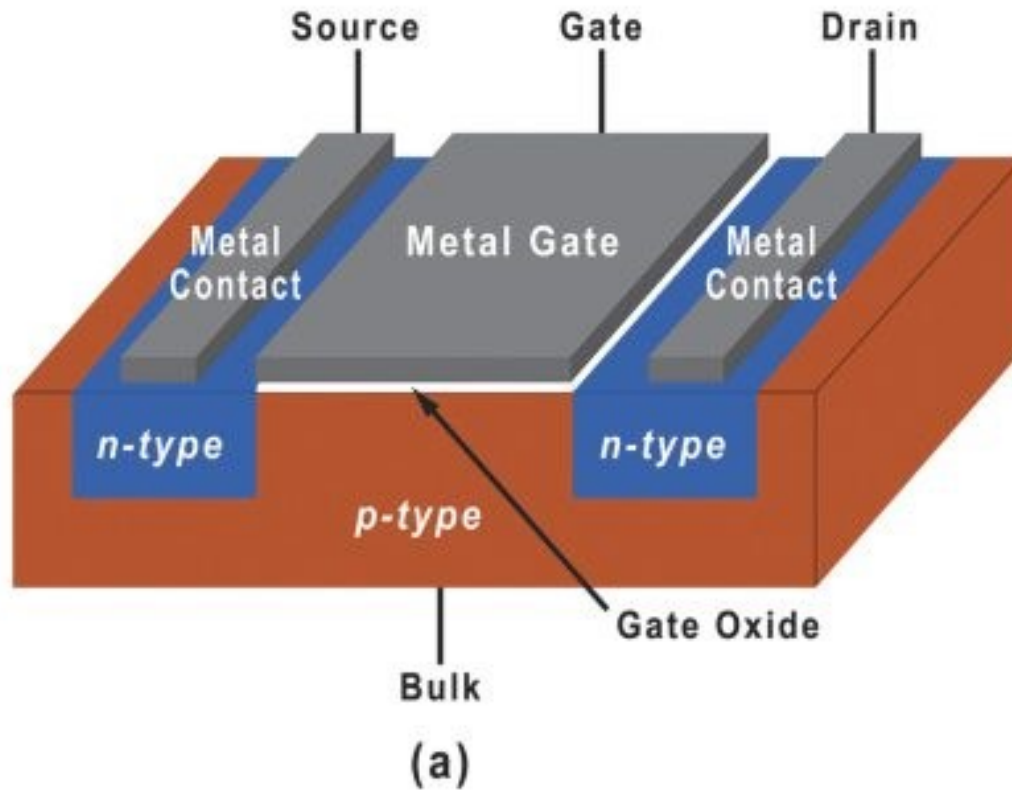


(f)
**PMOS used in
Digital circuits**

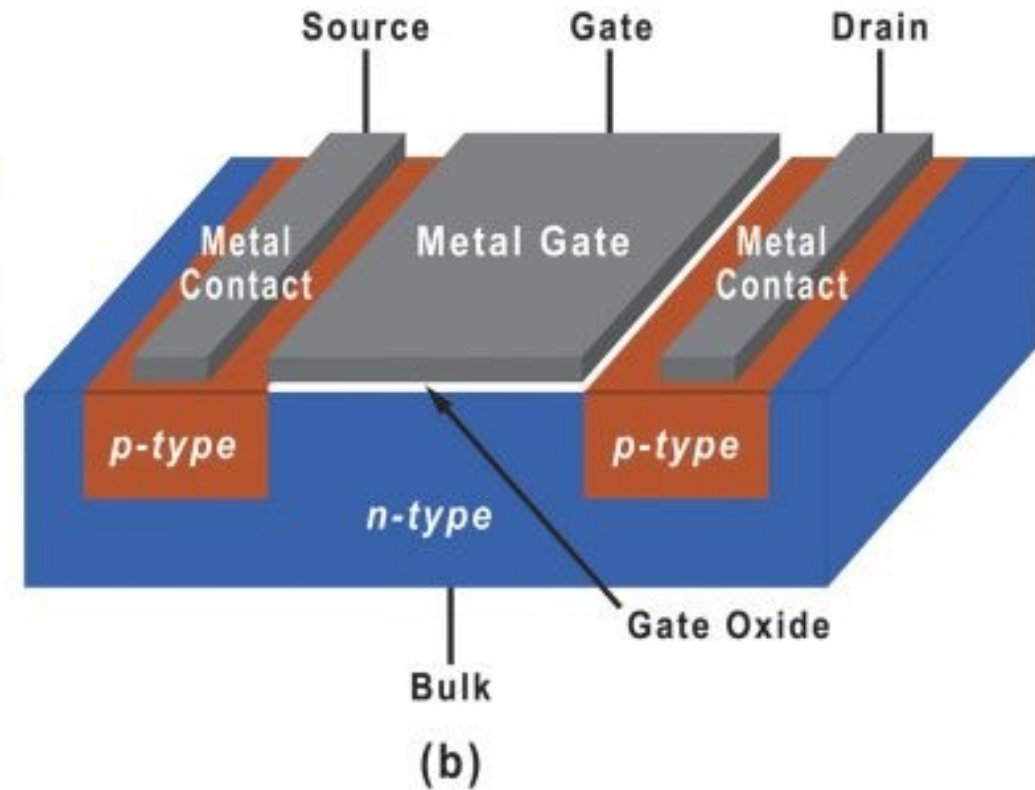
<https://www.electronics-tutorial.net/wp-content/uploads/2015/05/MOS-transistors.png>

MOS Transistor Types: nMOS & pMOS

(a): nMOS



(b): pMOS

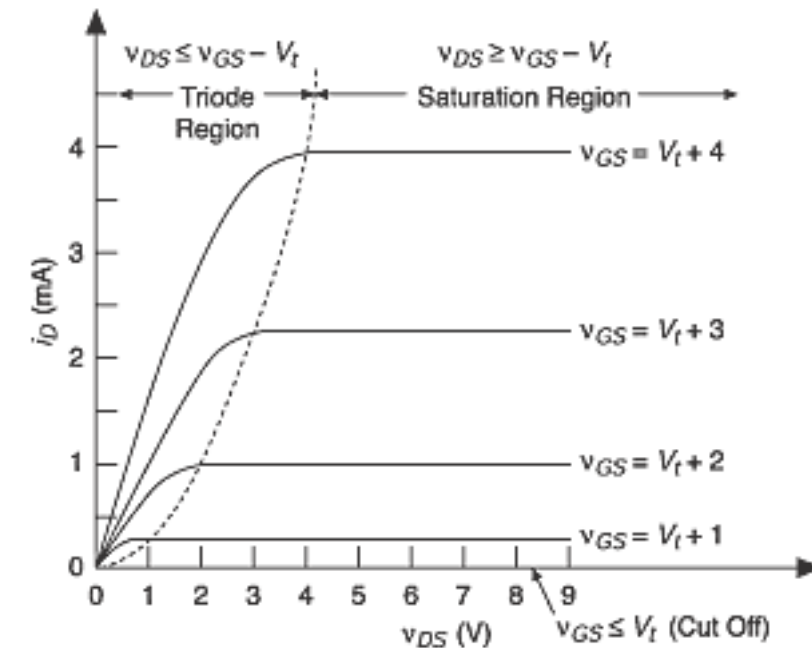
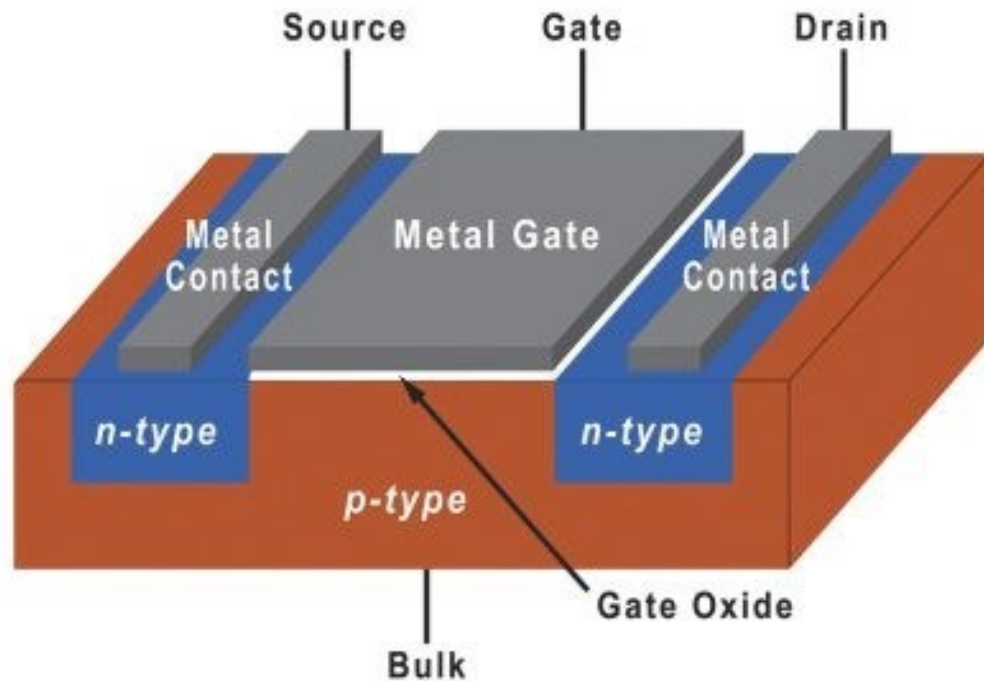


https://www.mks.com/mam/celum/celum_assets/Figure_11-Semiconductor_Handbook_800w.jpg

MOS Transistor Types: nMOS

$V_{gs} < V_t$
 $V_{ds} < V_{gs} - V_t$ AND $V_{gs} > V_t$
 $V_{ds} > V_{gs} - V_t$ AND $V_{gs} > V_t$

OFF
Linear (Triode) region
Saturation



https://www.mks.com/mam/celum/celum_assets/Figure_11-Semiconductor_Handbook_800w.jpg

<https://allthingsvlsi.wordpress.com/wp-content/uploads/2013/04/nmos-char.gif?w=400&h=296>

MOS Transistor Types: pMOS

$$|V_{gs}| < |V_t| \quad (V_{sg} < |V_t|)$$

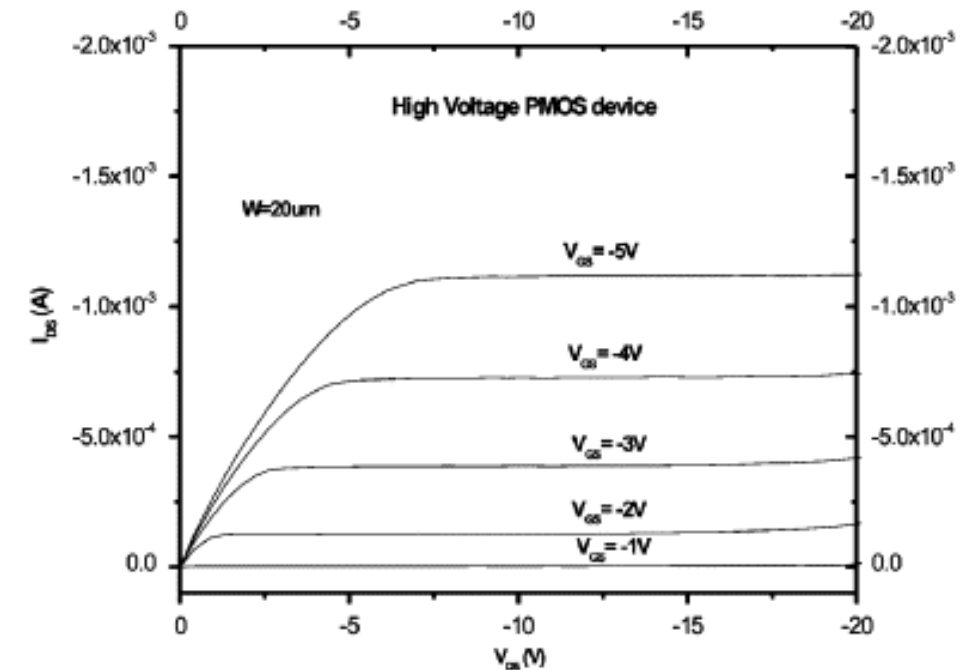
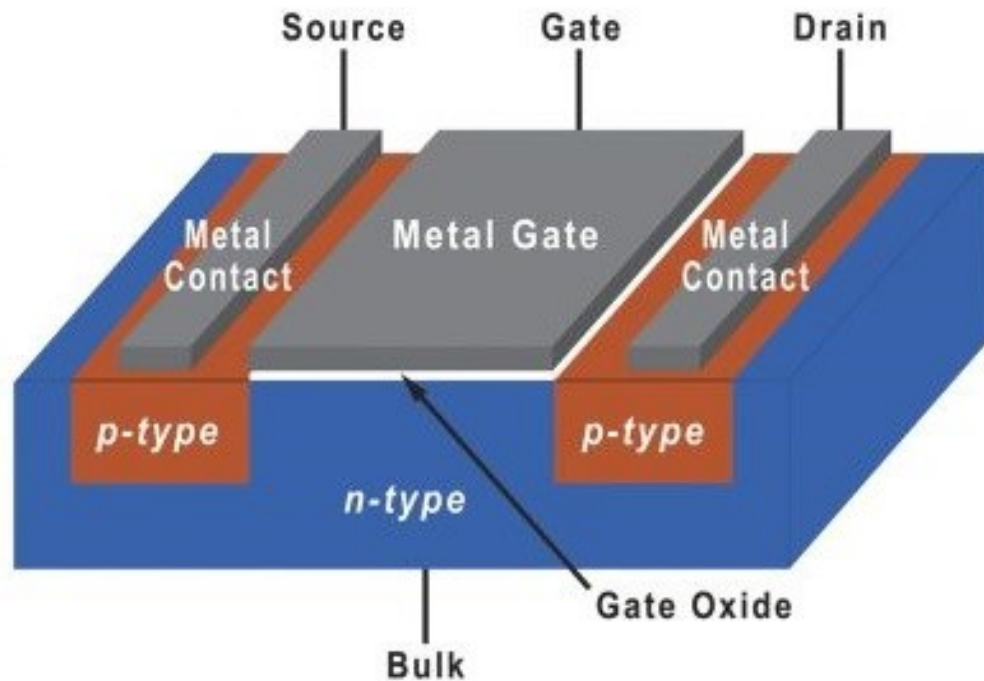
$$|V_{ds}| < |V_{gs}| - |V_t| \quad (V_{sd} < V_{gs} - |V_t|) \text{ AND } |V_{gs}| > |V_t|$$

$$|V_{ds}| > |V_{gs}| - |V_t| \quad (V_{sd} > V_{gs} - |V_t|) \text{ AND } |V_{gs}| > |V_t|$$

OFF

Linear (Triode) region

Saturation



https://www.mks.com/mam/celum/celum_assets/Figure_11-Semiconductor_Handbook_800w.jpg

<https://allthingsvlsi.wordpress.com/wp-content/uploads/2013/04/pmos-char.gif?w=400&h=260>